

Report
Of
SOFTWARE ENGINEERING AND WEB DEVELOPMENT

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: CS-2341738

Name of Student: Paramjibi Sutar

Batch: 2023-27

2026-27

CANDIDATE'S DECLARATION

I, Shakyasimha Das do hereby declare that the internship report titled “**Software Engineering and Web Development**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

A handwritten signature in blue ink, appearing to read 'P. Sutar', is shown within a rectangular frame.

Signature

Student Name : Paramjibi Sutar

Roll No. : CS-2341738

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

A handwritten signature in blue ink, appearing to read 'Paramjibi Sutar', with a stylized flourish above the name.

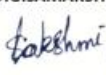
Signature

Student Name : Paramjibi Sutar

Roll No. : CS-2341738

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3. Internship Completion Certificate

Date: 19th August 2026 Employee ID: DIGI/DI/2026/0222	 Reference No.: DIGI/IOL/0897
INTERNSHIP COMPLETION CERTIFICATE TO WHOMSOEVER IT MAY CONCERN	
This is to certify that Paramjibi Sutar, s/d/o Mr.Kshirod Sutar, has successfully completed an internship as UI/UX Intern in the Technology & Development Department at DIGISAMAKSH Private Limited. The internship tenure was from June 01, 2026 to August 15, 2026. During the internship period, Paramjibi Sutar has shown a high level of commitment, technical understanding, and problem-solving ability while working on assigned projects and tasks.	
Key Responsibilities & Contributions ✦ Conducting user research and translating findings into wireframes and low-fidelity prototypes. ✦ Designing high-fidelity visual interfaces using tools such as Figma, Adobe XD, or Sketch. ✦ Collaborating with the Development team to ensure design feasibility and pixel-perfect implementation. ✦ Assisting in the creation and maintenance of a comprehensive Design System and UI Kits. ✦ Iterating on designs based on user feedback and usability testing results.	
Internship Evaluation Based on the overall performance, learning attitude, and contributions during the internship, Paramjibi Sutar has been evaluated on the following parameters: ✦ Work Quality: Delivered assigned tasks with accuracy and within timelines. ✦ Communication Skills: Maintained professional communication and clarity while coordinating with team members. ✦ Learning & Adaptability: Showed eagerness to learn new skills and adapt to evolving project requirements. ✦ Professional Conduct: Maintained discipline, punctuality, and a positive approach throughout the internship. Overall, Paramjibi Sutar has successfully met the expectations of the internship program and has added value to the Technology & Development Department of DIGISAMAKSH Private Limited. We wish Paramjibi Sutar success in all future endeavors.	
For DIGISAMAKSH Private Limited   Lakshmi Raghavan Director – HR +91 9686634464	For DIGISAMAKSH Private Limited   Atul Garg Director – Growth +91 7303767575
DIGISAMAKSH Private Limited Registered Office: D 180, Alpha 1, Greater Noida, UP – 201306 CIN: U80210UP2022PTC175156; Web: www.digisamaksh.com; Email: digisamaksh@gmail.com; ☎: 91+(0)9582349641; in/company/digisamaksh; fb/digisamaksh 	

4. Project Description

4.1 Introduction

The internship program provided an opportunity to gain practical exposure to the field of **Software Engineering and Web Development** (UI/UX Intern) in a professional working environment. The internship was undertaken at **DIGISAMAKSH Private Limited** as a **Software Engineering Intern** in the Technology & Development Department from **01 June 2026 to 15 August 2026**.

During the internship, the work involved gaining hands-on experience in various aspects of web and software development, including frontend and backend development, website development and maintenance, database-related activities, software testing, debugging, troubleshooting, code review, and technical documentation. The internship also provided an opportunity to understand how software development tasks are performed collaboratively within a professional team.

The internship helped in developing practical technical skills while strengthening problem-solving, debugging, teamwork, communication, and professional development practices. It also provided valuable exposure to the application of concepts learned during the B.Tech CSE program in real-world software development activities.

Due to the **Non-Disclosure Agreement (NDA)** signed with the organization, confidential information such as proprietary source code, client information, internal system architecture, technical specifications, private datasets, and other sensitive business information has not been included in this report. The report therefore presents the internship experience and technical work at a generalized academic level while respecting the organization's confidentiality requirements.

4.2 Organization Profile

DIGISAMAKSH Private Limited is an organization working in different areas such as **human resources consulting, digital marketing, software development, e-commerce solutions, data analytics, and strategic consultancy services**. The company provides services and solutions according to the requirements of its clients.

The organization also has a **Technology & Development Department**, where software and web development activities are carried out. These activities include developing and maintaining websites and software applications, database-related work, testing, debugging, and technical documentation.

During my internship, I worked as a **Software Engineering Intern** in the Technology & Development Department. The internship provided me with an opportunity to learn about professional software development practices and gain practical experience in web and software development.

Due to the **Non-Disclosure Agreement (NDA)**, confidential information related to the company's clients, source code, internal systems, technical specifications, and proprietary processes has not been included in this report.

4.3 Problem Statement

During the internship, the main focus was to gain practical experience in **software and web development** while supporting the Technology & Development Department in its assigned tasks. The work involved understanding development requirements, assisting with coding, debugging and testing, maintaining existing systems, and preparing technical documentation.

One of the key challenges in a software development environment is to ensure that applications and systems function correctly, remain maintainable, and meet the

required project guidelines. Technical issues may arise during development or maintenance and need to be identified and resolved systematically.

Therefore, the internship provided an opportunity to understand and contribute to software development activities while developing practical skills in **coding, debugging, testing, system maintenance, documentation, research, and problem-solving**.

Due to the NDA signed with the organization, specific project details, proprietary implementation methods, source code, internal architecture, client information, and other confidential information are not disclosed in this report.

This version is based directly on the responsibilities and contributions mentioned in your **Internship Completion Certificate**, especially software development support, troubleshooting, system maintenance, documentation, research, and problem-solving.

4.4 Project Objectives

The main objectives of the internship were:

1. **To gain practical experience** in software and web development in a professional environment.
2. **To improve programming skills** by assisting in coding and software development activities.
3. **To develop debugging and testing skills** for identifying and resolving technical issues.
4. **To understand system maintenance** and support the smooth functioning of software applications.
5. **To improve technical documentation skills** by preparing and maintaining project-related documents.

6. **To research and learn new technologies, frameworks, and tools** used in software development.
7. **To develop problem-solving skills** by working on technical challenges and suggesting improvements.
8. **To improve teamwork and communication skills** by collaborating with developers, testers, and project team members.

These objectives are aligned with the responsibilities and contributions listed in your internship offer letter and completion certificate.

4.5 Scope of the Project

The scope of the internship was focused on gaining practical experience in **software engineering and web development** within the Technology & Development Department. The major areas covered during the internship were:

1. **Software Development:** Assisting in coding and development activities for assigned software tasks.
2. **Web Development:** Gaining practical exposure to frontend and backend development activities.
3. **Testing and Debugging:** Identifying technical issues, testing software applications, and assisting in resolving errors.
4. **System Maintenance:** Supporting the maintenance and smooth functioning of existing technical systems.
5. **Technical Documentation:** Preparing and maintaining project-related documentation, including code comments, user guides, and technical reports.
6. **Research and Learning:** Exploring new technologies, frameworks, and tools to support development activities.
7. **Team Collaboration:** Working with developers, testers, and project team members to complete assigned tasks and meet project requirements.

8. **Problem-Solving and Improvement:** Understanding technical challenges and contributing suggestions for improving workflows and project outcomes.

The scope described above is based on the responsibilities and contributions stated in the **offer letter and internship completion certificate**. Specific project implementations, source code, internal architecture, client information, and proprietary processes are excluded from this report in accordance with the NDA.

4.6 Technologies and Tools Used

During the internship, I worked primarily on **backend development and Artificial Intelligence (AI)**. The main technologies and tools used during the internship were:

1. **C#** – Used for backend programming and development.
2. **.NET** – Used for developing backend applications and implementing server-side functionality.
3. **PostgreSQL** – Used for database management, storing, retrieving, and managing application data.
4. **Artificial Intelligence (AI)** – Used as part of the development work involving AI-based functionality and solutions.

These technologies provided practical experience in backend development, database management, and AI-related software development.

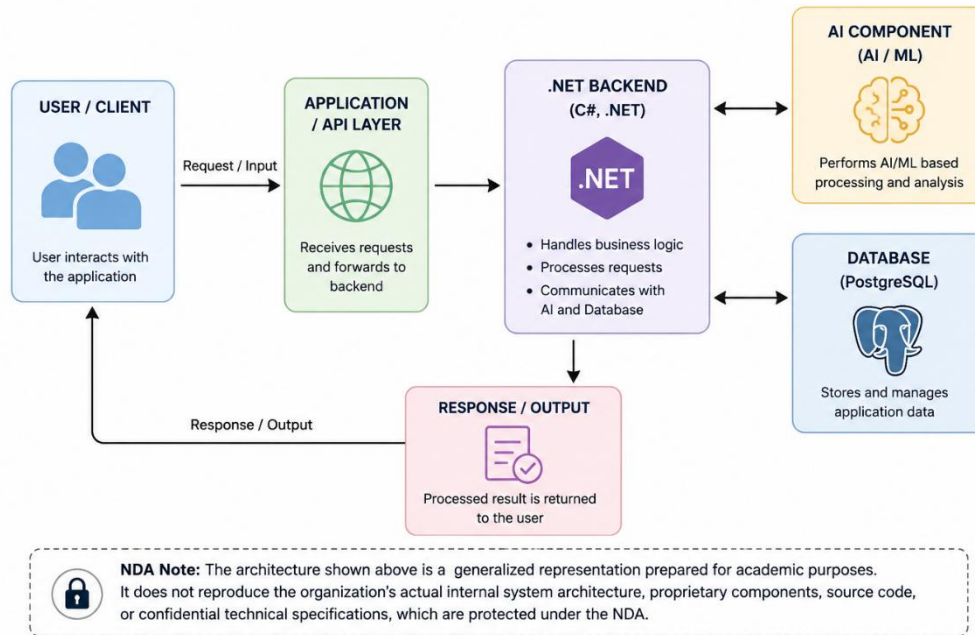
4.7 System Architecture

The system architecture used during the internship can be represented as a **generalized backend and AI-based architecture**. Since the internship work was covered under an NDA, specific internal architecture and proprietary implementation details are not included.

The overall architecture consists of the following major components:

GENERALIZED SYSTEM ARCHITECTURE

(For Academic Purpose Only)



Description of the Architecture

- **User/Client:** Provides requests or inputs to the application.
- **Application/API:** Receives the requests and communicates with the backend.
- **.NET Backend:** Handles the main application logic, request processing, and communication between different components.
- **AI Component:** Handles the AI-related functionality required by the application.
- **PostgreSQL Database:** Stores and manages the required application data.
- **Response/Output:** The processed result is returned to the user or client through the application.

This architecture provides a basic separation between the **application interface**, **backend logic**, **AI functionality**, and **database layer**, making the system easier to maintain and develop.

4.8 Methodology

The methodology followed during the internship was based on a systematic approach to software development and problem-solving. The major steps followed were:

1. **Requirement Understanding**

Understanding the assigned tasks, requirements, and expected outcomes before starting the development work.

2. **Planning and Analysis**

Analyzing the requirements and planning the appropriate approach for implementing the assigned tasks.

3. **Backend Development**

Working on backend development using **C# and .NET** to implement application logic and required functionalities.

4. **AI Integration and Development**

Working with **Artificial Intelligence (AI)** concepts and implementing AI-related functionality as required for the assigned work.

5. **Database Management**

Using **PostgreSQL** for storing, retrieving, and managing application data.

6. **Testing and Debugging**

Testing the developed functionality, identifying errors, and debugging the application to improve its reliability.

7. **System Maintenance**

Supporting existing systems by troubleshooting technical issues and assisting with necessary updates and improvements.

8. **Documentation**

Preparing and maintaining technical documentation related to the assigned work.

9. **Team Collaboration**

Working with developers, testers, and other team members to discuss requirements, solve problems, and complete assigned tasks.

10. Review and Improvement

Reviewing the completed work and making necessary improvements based on feedback and project requirements.

The methodology described above represents the internship activities at a **general academic level**. Specific internal development processes, proprietary methodologies, source code, system architecture, and confidential implementation details have been excluded in accordance with the NDA.

4.9 Expected Outcomes

The internship was expected to provide practical knowledge and experience in **Software Engineering, Backend Development, and Artificial Intelligence**. The major expected outcomes were:

1. Improved Backend Development Skills

To gain practical experience in backend development using **C# and .NET**.

2. Understanding of AI-Based Development

To gain practical exposure to **Artificial Intelligence** and its use in software development.

3. Database Management Skills

To develop an understanding of storing, retrieving, and managing data using **PostgreSQL**.

4. Improved Debugging and Testing Skills

To learn how to identify errors, test software functionality, and resolve technical issues.

5. System Maintenance Experience

To understand the basic practices involved in maintaining and troubleshooting software systems.

6. Technical Documentation Skills

To improve the ability to prepare and maintain technical documents and reports.

7. Problem-Solving Skills

To improve analytical thinking and develop effective solutions to technical challenges.

8. Teamwork and Communication

To gain experience working with developers, testers, and other team members in a professional environment.

9. Professional Experience

To understand professional software development practices and apply theoretical knowledge gained during the B.Tech CSE program in a practical environment.

These outcomes are consistent with the responsibilities and evaluation areas mentioned in your internship completion certificate, including software development support, research and learning, system maintenance, technical documentation, team collaboration, technical skills, problem-solving, and professional conduct.

4.10 Certificates of completion and Communication Mail Proof



Atul Garg <atul@digisamaksh.com>
to me ▾

May 3, 2026, 5:23 PM ☆ 😊 ↶ ⋮

Dear Paramjibi Sutar,

Following your successful interview at the IILM University Job Fair, we are pleased to officially offer you an internship position with DigiSamaksh.

Please note that your formal offer letter was dispatched to your University Coordinator approximately 20 days ago. To ensure you have immediate access to the details of your role, responsibilities, and terms of engagement, we have provided a digital copy below:

<https://drive.google.com/file/d/1hzcMUswvtXxmVKWXA38tDyVWUnY14jOm/view?usp=drivesdk>

Next Steps for Onboarding

To confirm your interest and formally accept this internship, please follow these steps:

1. **Review the Document:** Carefully read through the linked offer letter regarding your start date and internship duration.
2. **Email Confirmation:** Reply to this email with a formal statement of acceptance if you wish to join our team.
3. **Deadline:** As the initial offer was sent 20 days ago, please provide your response at your earliest convenience to secure your spot.

We were impressed by your profile during the job fair and believe you will be a valuable addition to our upcoming projects. We look forward to the possibility of working with you.

Best Regards

Lakshmi Raghavan

Director-HR, Digisamaksh(P)Ltd

M.No: [+91-9686634464](tel:+91-9686634464)

Date: 17th April 2026
Reference No.: DIGI/IOL/0897



OFFER LETTER TO WHOMSOEVER IT MAY CONCERN

To,
Paramjibi Sutar,
IILM University, Greater Noida, India

Dear **Paramjibi Sutar,**

We are delighted to extend an offer of internship to you for the position of UI/UX Design Intern in Design & Development Department at DIGISAMAKSH. Your exceptional qualifications and passion for Design & Development make you an ideal candidate for this internship opportunity.

We are confident that this internship will provide you with valuable hands-on experience and the opportunity to further develop your skills in UI/UX Design. In this role, your primary focus will be on the end-to-end user experience of DigiSamaksh's digital products. Your duties will include, but are not limited to:

- Conducting user research and translating findings into wireframes and low-fidelity prototypes.
- Designing high-fidelity visual interfaces using tools such as Figma, Adobe XD, or Sketch.
- Collaborating with the Development team to ensure design feasibility and pixel-perfect implementation.
- Assisting in the creation and maintenance of a comprehensive Design System and UI Kits.
- Iterating on designs based on user feedback and usability testing results.

Internship Details:

- Start Date: June 01, 2026
- End Date: August 15, 2026
- Working Hours: 6-8 hours in a day
- Location: Greater Noida - Hybrid (Office & Work from Home)

Compensation and Benefits:

As an intern, you will be eligible for the following compensation and benefits:

- Stipend: Unpaid
- Travel Reimbursement: If applicable
- Mentorship and Guidance from experienced Design professionals.
- Opportunity to attend Design-related workshops and training sessions.
- Certificate of Internship upon successful completion.
- Certificate of Appreciation as per performance
- Incentive can be provided as per company decision

Please note that this offer is contingent upon your acceptance of our terms and conditions, as well as satisfactory completion of any necessary background checks or paperwork required by our company policies.

To accept this offer, please sign and return a copy of this letter by April 25, 2026 indicating your agreement to the terms and conditions outlined herein. If you have any questions or require further information, please do not hesitate to contact me.

We are excited to have you join our team as a UI/UX Design Intern at DIGISAMAKSH. We believe that your enthusiasm, knowledge, and dedication will make a significant contribution to our organization. We look forward to working with you and wish you a successful internship experience.

For
DIGISAMAKSH Private Limited


Ms. Lakshmi Raghavan
Director – HR,
DigiSamaksh Private Limited
+91- 9686634464

DIGISAMAKSH Private Limited

Registered Office: D 180, Alpha 1, Greater Noida, UP – 201306

CIN: U80210UP2022PTC175156; Web: www.digisamaksh.com; Email: digisamaksh@gmail.com;

☎: 91+(0)9582349641; [in/company/digisamaksh](https://www.linkedin.com/company/digisamaksh); [ig/digisamaksh](https://www.instagram.com/digisamaksh); [fb/digisamaksh](https://www.facebook.com/digisamaksh)



Lakshmi Raghavan

<hr@digisamaksh.com>

to me

Wed, May 20, 2:02 PM

☆ 😊 ↶ ⋮

Dear Paramjibi Sutar

Greetings from Digisamaksh Group!!!

Thank you for accepting the internship offer with Digisamaksh Group. Your common induction is scheduled on June 01, 2026 at 10:00AM(IST) by Google Meet.

The Joining link is as below:

Induction-Digisamaksh
Monday, 1 June - 10:00 – 11:00am
Time zone: Asia/Kolkata
Google Meet joining info
Video call link: <https://meet.google.com/uxr-zpke-osv>
Or dial: (US) +1 401-719-1437 PIN: 776 543 760#
More phone numbers: <https://tel.meet/uxr-zpke-osv?pin=3048121894566>

It's Mandatory to join the Induction program on the given date & time.
For any doubt, feel free to contact the undersigned.

Best Regards,
Lakshmi Raghavan
HR Head, Digisamaksh Group
Mb No: [+91-9686634464](tel:+919686634464)

Offer Letter || Digisamaksh

Lakshmi Raghavan

<hr@digisamaksh.com>

to me

Thu, May 28, 2:37 PM

☆ 😊 ↶ ⋮

Dear Paramjibi Sutar

Congratulations!

We are pleased to extend an offer to you for the position of UI/UX Design Intern at Design & Development Department with DIGISAMAKSH Group based on your application and interview.

Please see the attached Offer Letter. Kindly reply to this email with the word "Accepted" by May 30, 2026, 8:00 PM IST. Otherwise, the Internship may be withdrawn at the discretion of the company.

Offer Letter: <https://drive.google.com/file/d/1hzcMU5nwtXmvtKWIA38tDVWUjYt4jOm/view?usp=drivesdk>

Kindly Note:

- The duration of the internship is as per the offer letter, with flexible working hours and an online mode.
- A stipend would be paid as per the offer for this internship.
- Performance-based incentives can be given during your internship.
- A completion letter for the internship would be given.
- If performance is good, an appreciation letter would be provided along with a letter of completion.
- Job Offers can be given if performance is extraordinary.
- Make sure you have a laptop/desktop and a good internet connection.

Once accepted, your Joining Meeting with Mentor will be scheduled.

Best Regards
Lakshmi Raghavan
Director-HR, Digisamaksh Group
[+91-9686634464](tel:+919686634464)

Lakshmi Raghavan

<hr@digisamaksh.com>

to me

Aug 20, 2026, 10:57 AM (10 days ago)

☆ 😊 ↶ ⋮

Dear Paramjibi Sutar

Greetings!!!!

We hope this email finds you in good health and high spirits. We are delighted to inform you that you have successfully completed your internship with DIGISAMAKSH. It gives us immense pleasure to acknowledge your hard work, dedication, and contributions during your time with us.

We would like to express our appreciation for your exceptional performance and the positive impact you have made during your internship. Your commitment to excellence, willingness to learn, and proactive approach have been commendable, and we believe that you have gained valuable skills and experience that will serve as a strong foundation for your future endeavors.

Please find attached the digital copy of your
Certificate of Internship: https://drive.google.com/file/d/1dXrAFR_b3u0yN113jKU8aTDMX8xAbA3-/view?usp=drivesdk

LOR : NA

We recommend that you download and save the certificate for future reference and include it in your portfolio to showcase your accomplishments.

We would like to take this opportunity to express our gratitude for your hard work, enthusiasm, and positive attitude. Your dedication to your assigned tasks and your eagerness to take on new challenges have made you a valuable member of our team. We have no doubt that you will continue to achieve great things in your academic and professional journey.

Once again, congratulations on successfully completing your internship. We wish you the very best in all your future endeavors. Should you ever require any assistance or references, please do not hesitate to reach out to us.

Best Regards,
HR Department
Digisamaksh(P)Ltd

5. Bibliography/References

- Microsoft, **C# Documentation**, Microsoft Learn.
- Microsoft, **.NET Documentation**, Microsoft Learn.
- PostgreSQL Global Development Group, **PostgreSQL Documentation**.
- Microsoft, **ASP.NET Core Documentation**, Microsoft Learn.
- Microsoft, **Artificial Intelligence and Machine Learning Documentation**, Microsoft Learn.
- DIGISAMAKSH Private Limited, **Internship Offer Letter**, 17 April 2026.
- DIGISAMAKSH Private Limited, **Internship Completion Certificate**, 19 August 2026.
- DIGISAMAKSH Private Limited, **Non-Disclosure Agreement**, 01 June 2026.